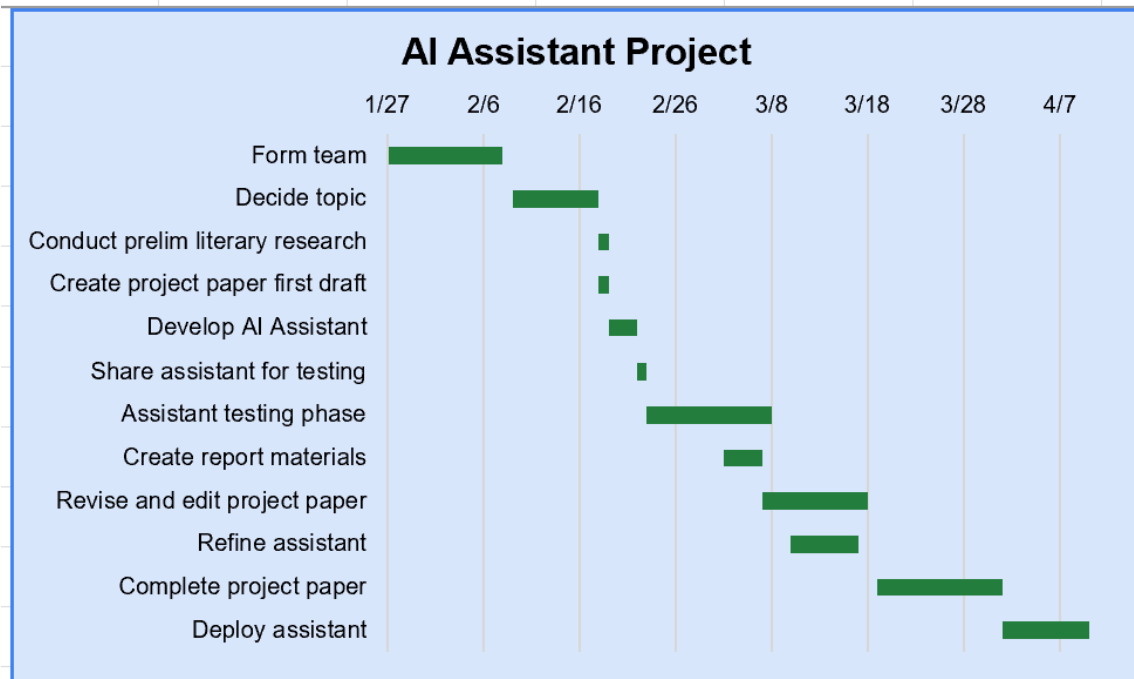


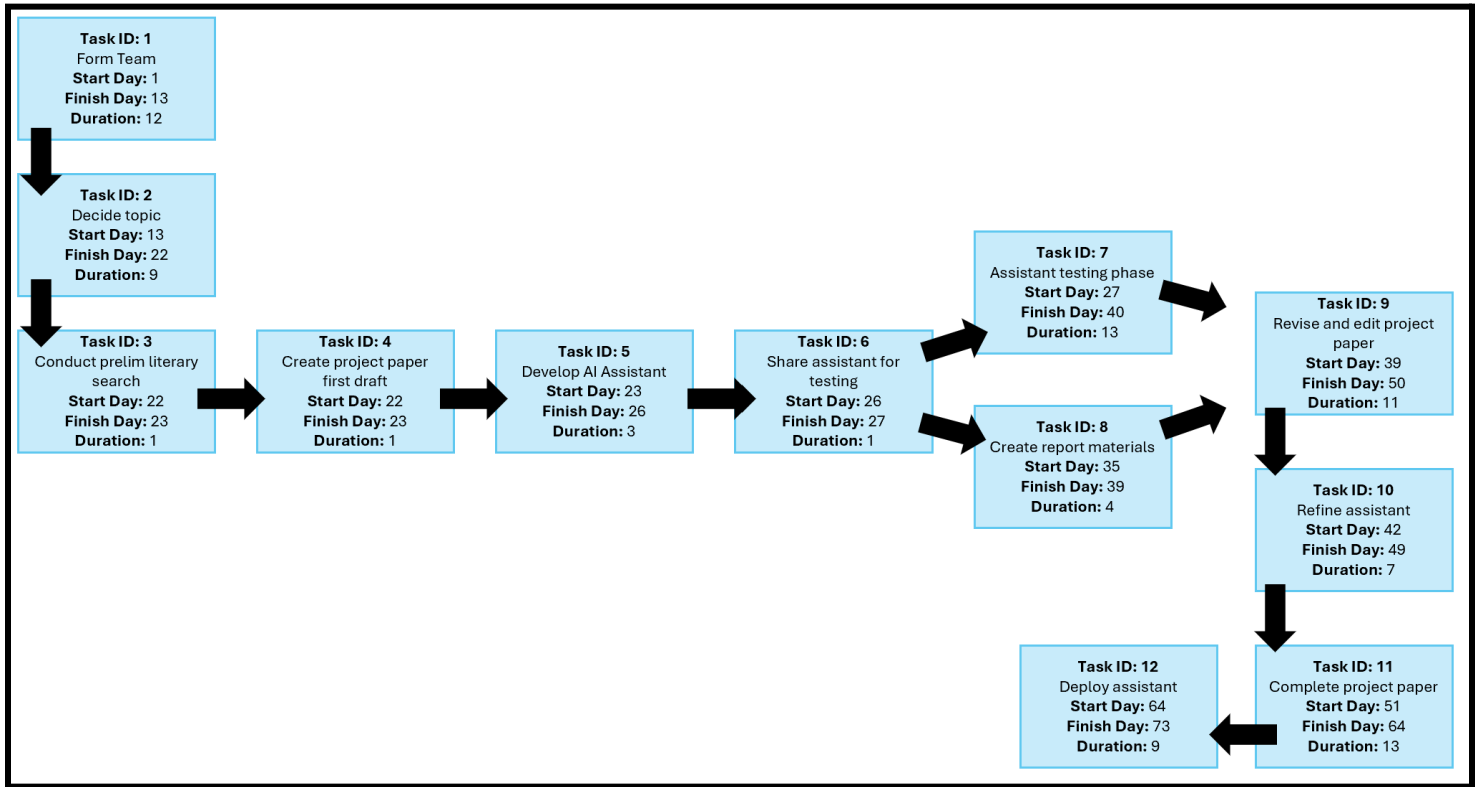
Phase 1 Report

AI HAIR ASSISTANT PROJECT REPORT

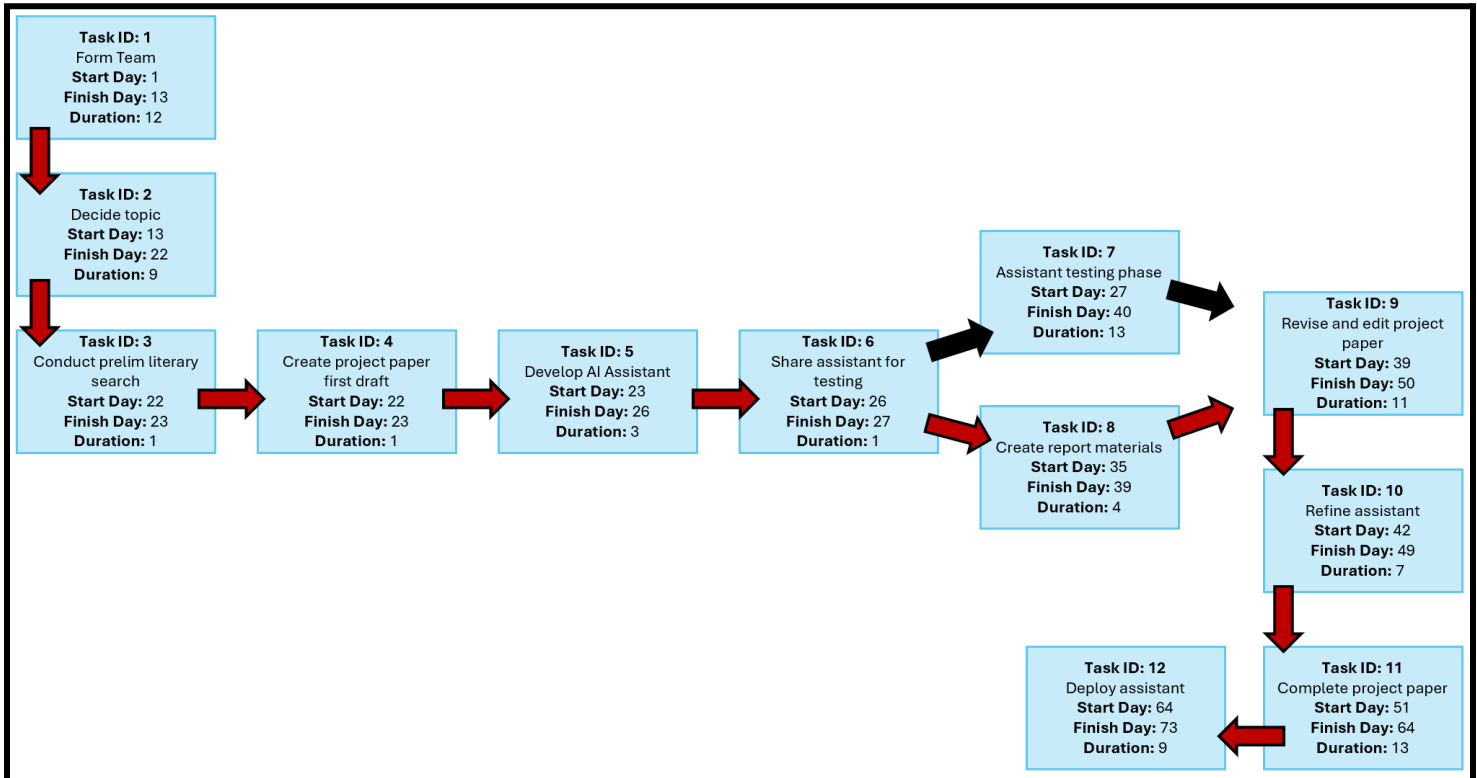
1. WORK BREAKDOWN STRUCTURE, GANTT CHART, AND PERT CHART

Task ID	Task Name	Start Date	Finish Date	Duration	Assigned To
1	Form team	1/27	2/8/2026	12	All
2	Decide topic	2/9	2/18/2026	9	All
3	Conduct prelim literary research	2/18	2/19/2026	1	Quarshona
4	Create project paper first draft	2/18	2/19/2026	1	Quarshona
5	Develop AI Assistant	2/19	2/22/2026	3	Quarshona
6	Share assistant for testing	2/22	2/23/2026	1	Quarshona
7	Assistant testing phase	2/23	3/8/2026	13	All
8	Create report materials	3/3	3/7/2026	4	Courtney
9	Revise and edit project paper	3/7	3/18/2026	11	All
10	Refine assistant	3/10	3/17/2026	7	Quarshona
11	Complete project paper	3/19	4/1/2026	13	All
12	Deploy assistant	4/1	4/10/2026	9	All



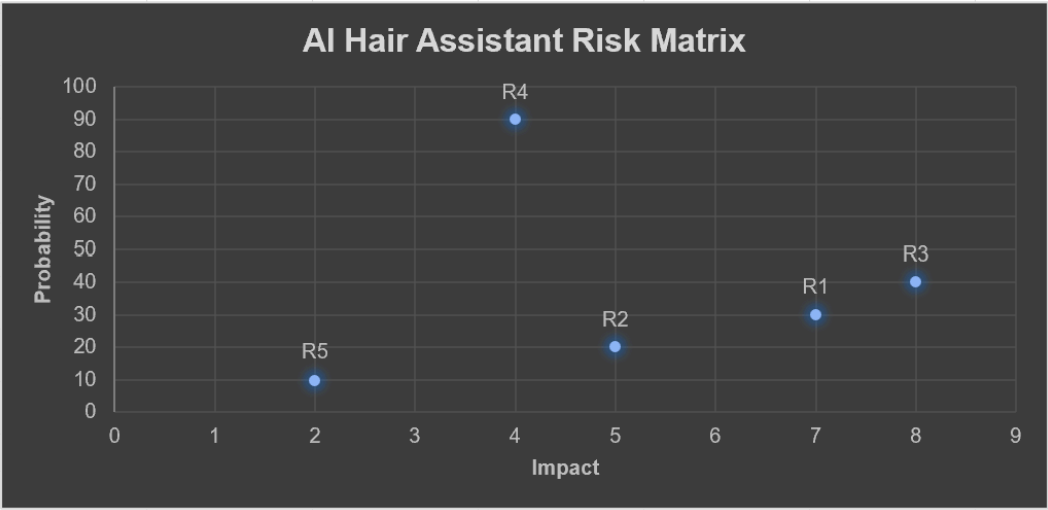


PERT CHART WITH CRITICAL PATH:

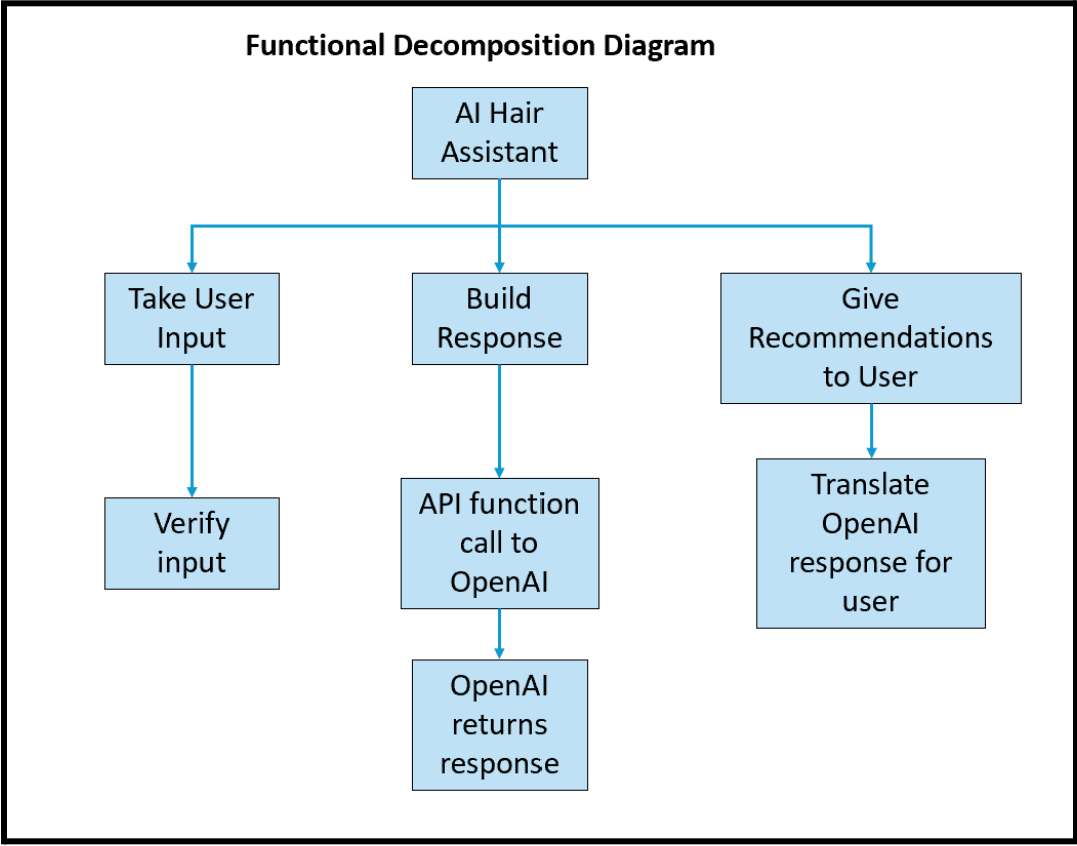


2. RISK MATRIX

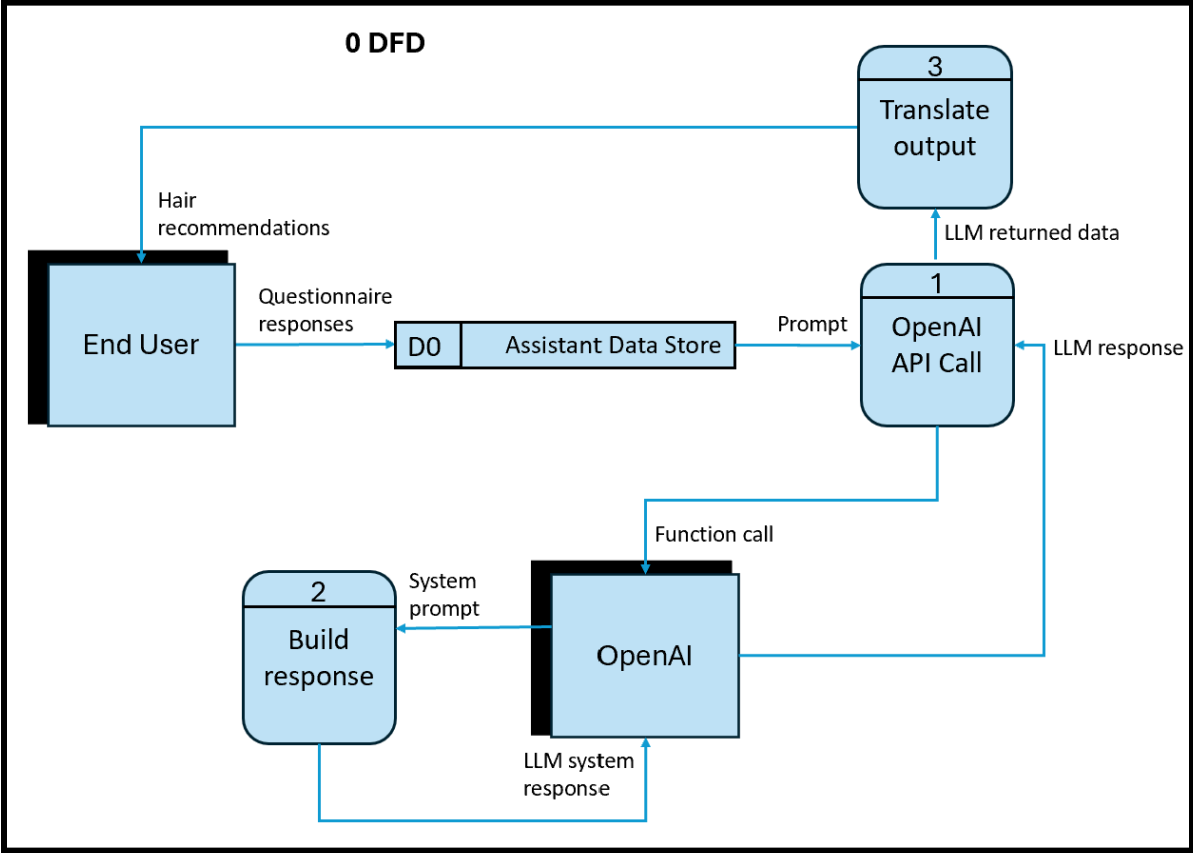
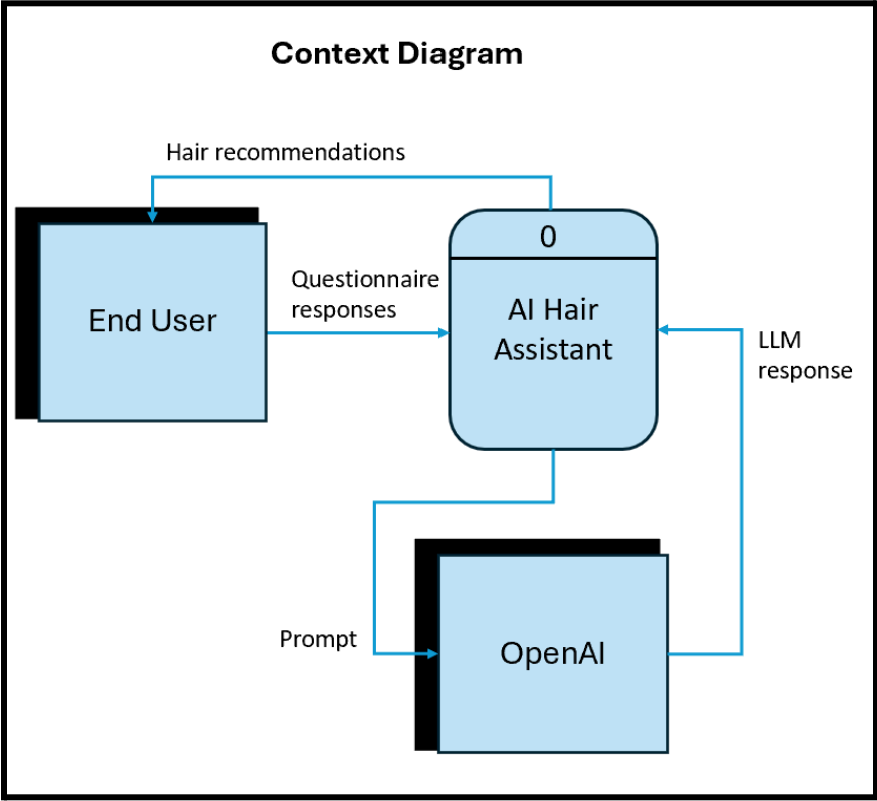
Risk ID	Risk Name	Impact	Probability
R1	Team member leaves group	7	30
R2	Unforeseen circumstances (natural disaster, political unrest, etc.)	5	20
R3	Assistant stuck in development stage	8	40
R4	Delays caused by outside responsibilities (work for other classes, families)	4	90
R5	OpenAI Secret Key made public	2	10

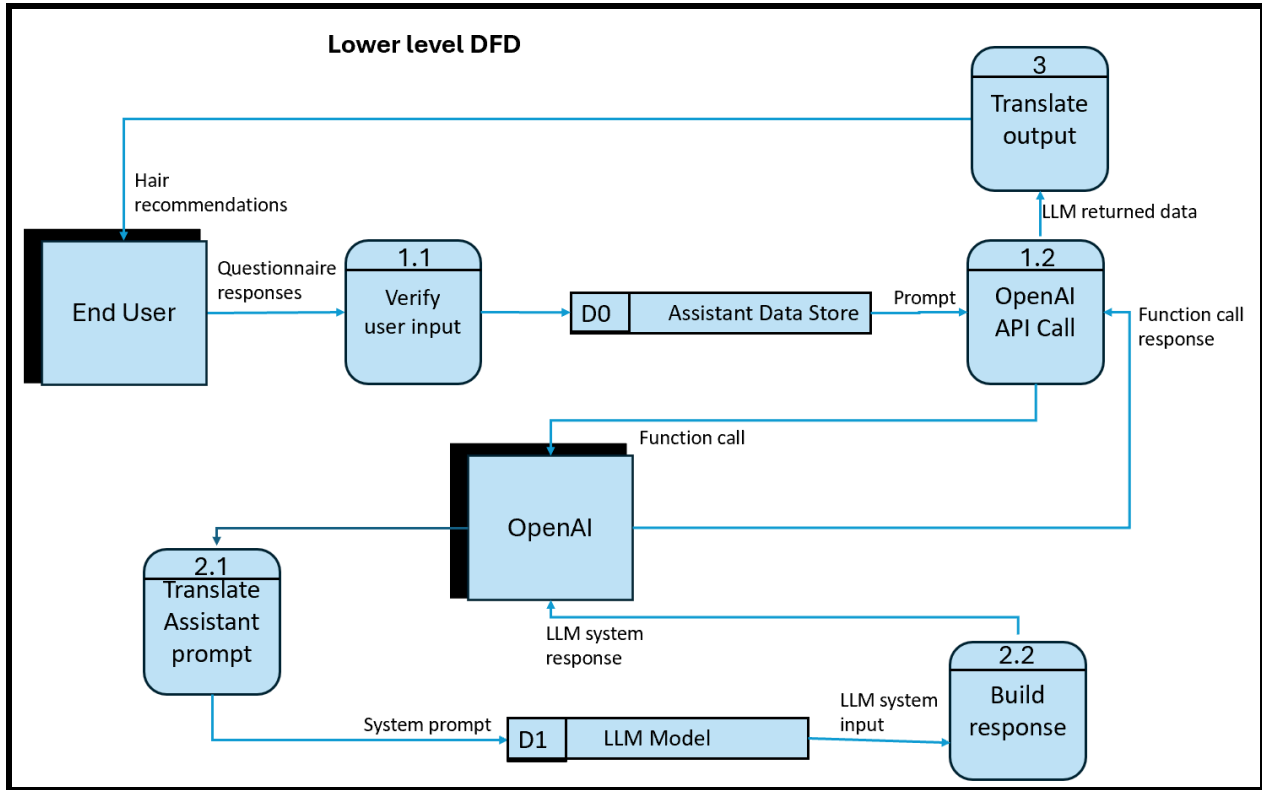


3. FUNCTIONAL DECOMPOSITION DIAGRAM



4. CONTEXT DIAGRAM, 0 DFD, AND LOWER LEVEL DFD





5. DECISION TABLE WITH 3 CONDITIONS

Verify User Input	1	2	3	4	5	6	7	8
Dropdown menus filled out	Y	-	-	Y	-	Y	N	-
Description typed in text box	-	Y	-	Y	-	-	Y	-
Input is about hair condition/problems	N	N	Y	Y	N	Y	Y	N
Accept input			X	X		X	X	
Reject input	X	X			X			X

6. CODE/PROTOTYPE

Prototype screenshots:

The screenshot shows a web browser window with the title "AI Hair Care Recommendation System". The address bar shows "localhost:8501". The page content includes a title "AI Hair Care Recommendation System", a subtitle "Fill out the questionnaire below to receive personalized hair care recommendations.", and four dropdown menus for "Hair Texture" (Straight), "Scalp Condition" (Dry), "Damage Level" (Low), and "Main Goal" (Moisture). Below these is a text input field for "Describe your main hair concerns".

AI Hair Care Recommendation System

Fill out the questionnaire below to receive personalized hair care recommendations.

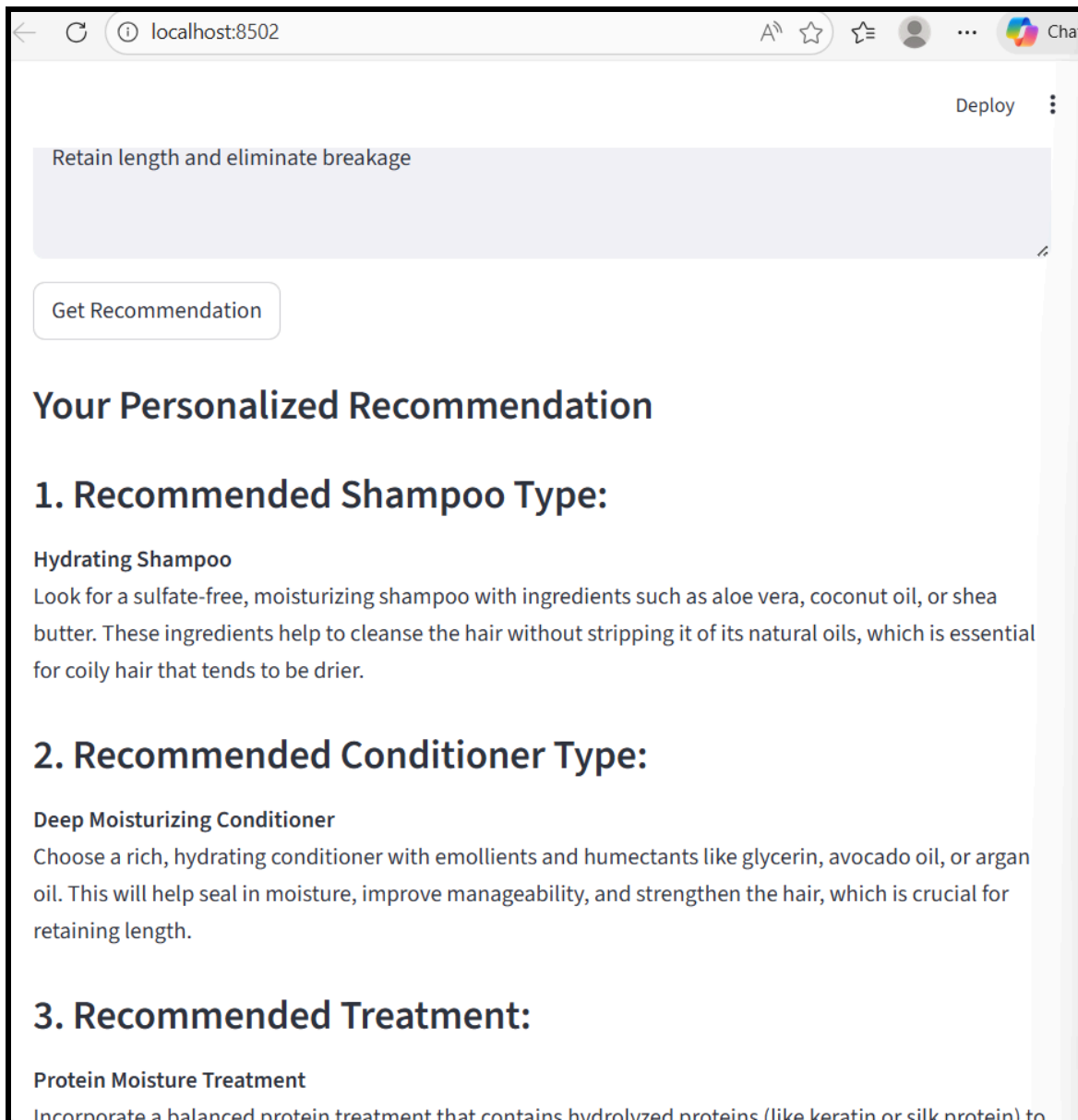
Hair Texture
Straight

Scalp Condition
Dry

Damage Level
Low

Main Goal
Moisture

Describe your main hair concerns



3. Recommended Treatment:

Protein Moisture Treatment

Incorporate a balanced protein treatment that contains hydrolyzed proteins (like keratin or silk protein) to help strengthen and reinforce hair structure while maintaining moisture. Look for treatments that add hydration alongside protein to avoid making the hair too stiff.

4. Suggested Weekly Routine:

- Day 1:
 - Wash with hydrating shampoo.
 - Condition with deep moisturizing conditioner.
 - Apply a leave-in conditioner for added moisture.
- Day 2:
 - Refresh curls with a water-based curl cream or spray mixed with a little leave-in conditioner.
- Day 3:
 - No wash; consider a scalp massage with light oil (like jojoba) to maintain scalp health.
- Day 4:
 - Use a protein moisture treatment as needed (once every 3-4 weeks, not on the same week as a deep conditioning treatment).

- Use a protein moisture treatment as needed (once every 3-4 weeks, not on the same week as a deep conditioning treatment).
- **Day 5:**
 - Co-wash with a cleansing conditioner (sulfate-free) to remove product build-up without stripping moisture.
- **Day 6/7:**
 - Apply a hydrating hair mask and leave on for a few hours or overnight, then rinse out and follow with conditioner.

5. Explanation for Recommendations:

Hair Type: Coily

Coily hair is typically dry and prone to breakage, making moisture retention critical. Hydrating shampoo and deep conditioners are essential to cleanse the hair without stripping natural oils.

Scalp Condition: Normal

With a normal scalp condition, you don't need specialized products for scalps dealing with oil or flakes, allowing focus on the hair's moisture needs.

Damage Level: Moderate

Since there is moderate damage, the goal should be a combination of strengthening and hydrating products. The recommended protein moisture treatment aids in reinforcing the hair structure while nourishing it.

Goal: Moisture, Retain Length, Eliminate Breakage

5. Explanation for Recommendations:

Hair Type: Coily

Coily hair is typically dry and prone to breakage, making moisture retention critical. Hydrating shampoo and deep conditioners are essential to cleanse the hair without stripping natural oils.

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With a normal scalp condition, you don't need specialized products for scalps dealing with oil or flakes, allowing focus on the hair's moisture needs.

Damage Level: Moderate

Since there is moderate damage, the goal should be a combination of strengthening and hydrating products. The recommended protein moisture treatment aids in reinforcing the hair structure while nourishing it.

Goal: Moisture, Retain Length, Eliminate Breakage

The routine emphasizes moisture with essential hydrating products, which are critical for coily hair. Regular protein treatments promote strength and minimize breakage, addressing your need to retain length.

Incorporating these recommendations into your hair care routine can significantly enhance moisture retention and mitigate breakage, supporting overall hair health and growth.

Streamlit DSS Hair Recommendation System

CODE :

```
import streamlit as st
import openai

st.title("AI Hair Care Recommendation System")
st.subheader("Decision Support System Prototype")

# ---- USER INPUT SECTION ----

st.header("Hair Assessment Survey")

hair_type = st.selectbox(
    "What is your hair type?",
    ["Straight", "Wavy", "Curly", "Coily"]
)

scalp_condition = st.selectbox(
    "How would you describe your scalp?",
    ["Dry", "Oily", "Normal", "Sensitive"]
)

hair_goal = st.selectbox(
    "What is your main hair goal?",
    ["Hair growth", "Moisture", "Damage repair", "Frizz control"]
)

heat_usage = st.selectbox(
    "How often do you use heat styling tools?",
    ["Never", "Sometimes", "Often"]
)

chemical_treatments = st.selectbox(
    "Do you use chemical treatments (relaxer, color, etc.)?",
    ["No", "Occasionally", "Frequently"]
)

# ---- DECISION SUPPORT ENGINE ----

if st.button("Get Recommendation"):
```

```
st.header("AI Hair Recommendation")
```

```
# Build AI prompt
```

```
prompt = f"""
```

```
A user completed a hair care survey.
```

```
Hair Type: {hair_type}
```

```
Scalp Condition: {scalp_condition}
```

```
Hair Goal: {hair_goal}
```

```
Heat Styling Frequency: {heat_usage}
```

```
Chemical Treatments: {chemical_treatments}
```

```
Based on these inputs, recommend:
```

- A hair care routine
 - Key ingredients to look for
 - Product types to use
 - Tips to improve hair health
- ```
"""
```

```
OpenAI API call
```

```
openai.api_key = "YOUR_API_KEY"
```

```
response = openai.ChatCompletion.create(
```

```
 model="gpt-4o-mini",
```

```
 messages=[
```

```
 {"role": "user", "content": prompt}
```

```
]
```

```
)
```

```
recommendation = response.choices[0].message.content
```

```
st.write(recommendation)
```

```
---- SYSTEM EXPLANATION SECTION ----
```

```
st.sidebar.title("System Architecture")
```

```
st.sidebar.write("""
```

```
This prototype demonstrates a Decision Support System.
```

```
Process:
```

- 1 User answers hair care survey
- 2 Streamlit collects inputs
- 3 Python builds a structured AI prompt
- 4 OpenAI model analyzes hair characteristics
- 5 AI generates personalized recommendations

This demonstrates how AI can support  
personalized beauty and wellness decisions.  
""")

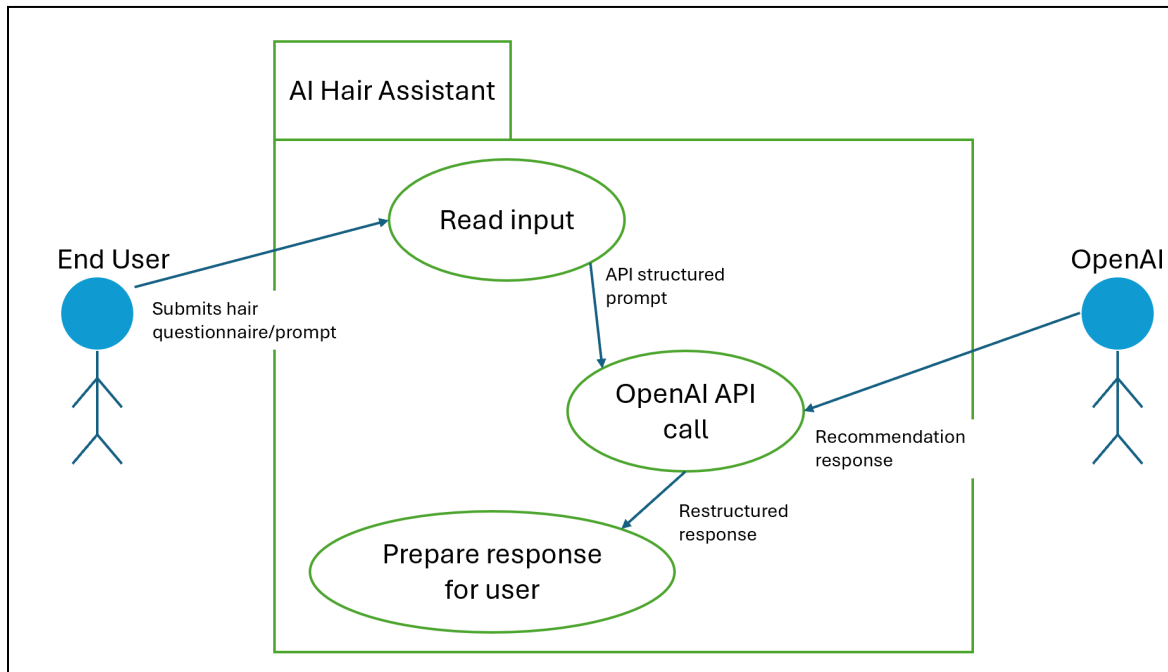
# Phase 2 Report



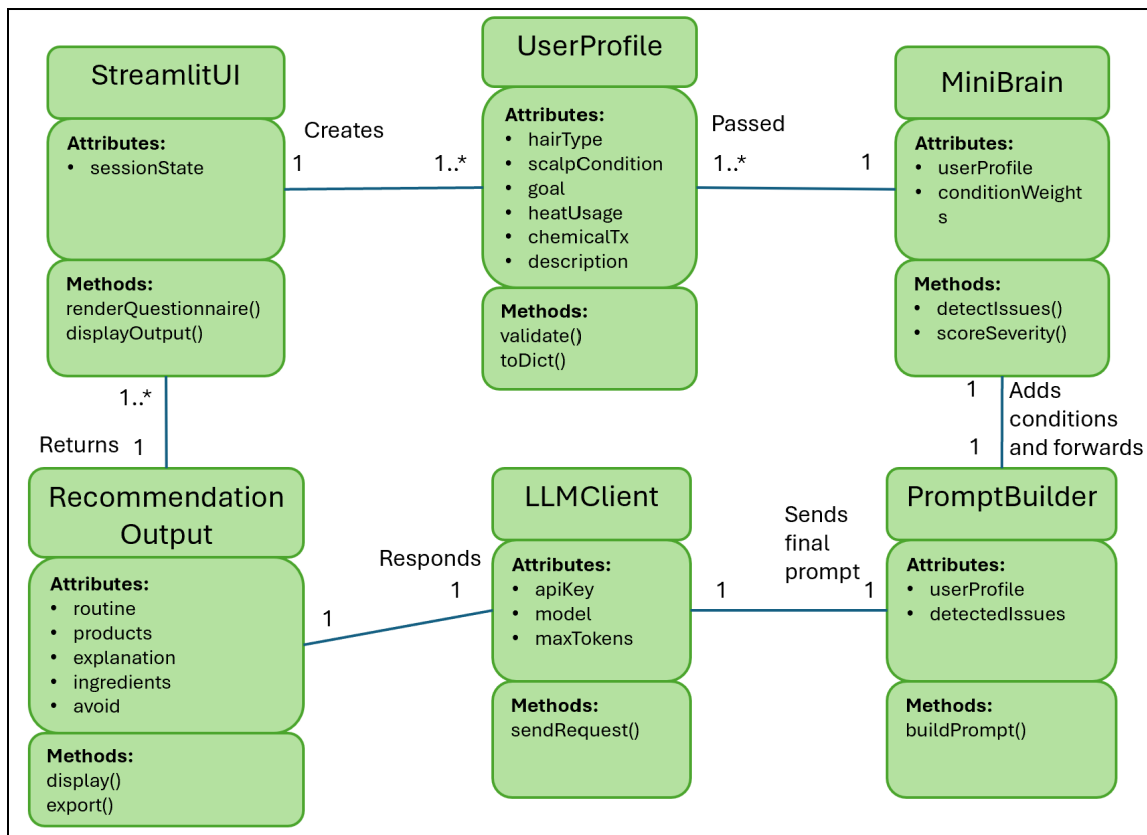
## PHASE 2 REPORT

### System Modeling

#### 1. Use Case Diagram



#### 2. Class Diagram



3. User Interface

# AI Hair Care Recommendation System

## Decision Support System Prototype

### Hair Assessment Survey

What is your hair type?

Straight

▼

How would you describe your scalp?

Dry

▼

What is your main hair goal?

Hair growth

▼

How often do you use heat styling tools?

Never

▼

What is your main hair goal?

Hair growth

How often do you use heat styling tools?

Never

Do you use chemical treatments (relaxer, color, etc.)?

No

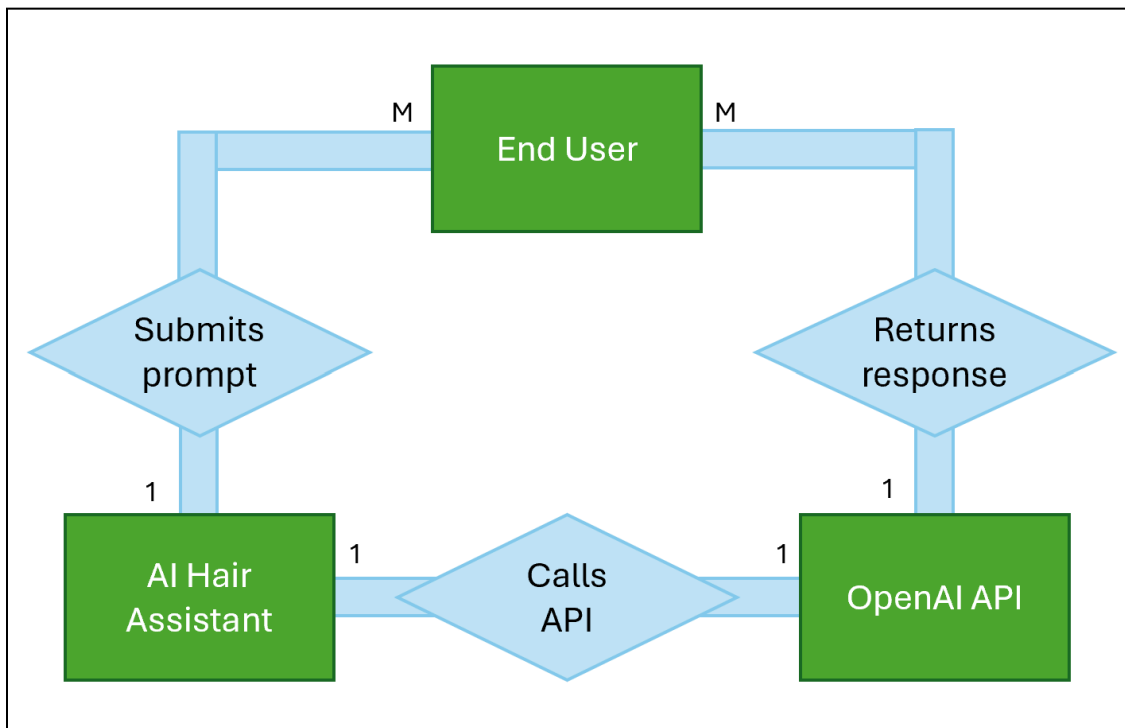
Tell us more about your hair (optional):

Example: My hair breaks easily at the ends and I want it longer...

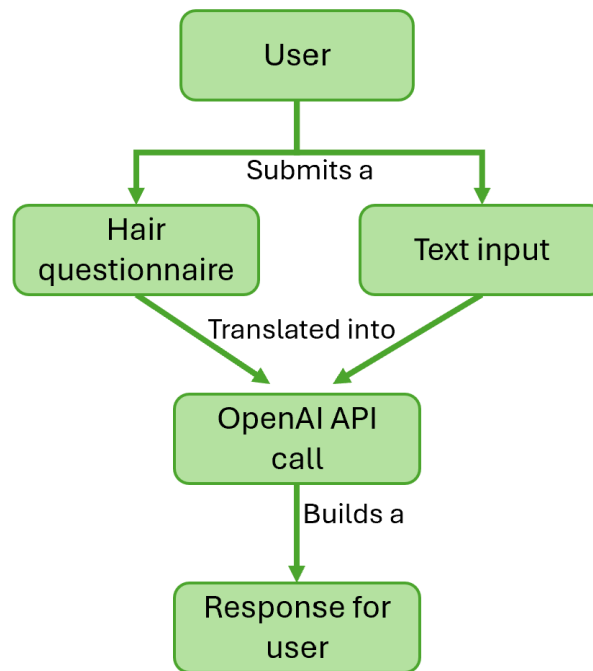
Get Recommendation

#### 4. E-R Diagram & Object Relationship Diagram

##### a. E-R Diagram



b. Object Relationship Diagram



## LLM Integration

1. Design and architecture (how does the system communication with LLM)

The AI Hair Assistant takes the input received from the user through text box or dropdown menus, then it restructures that input to build a prompt for OpenAI API. It informs the API that a user has completed a hair care survey, passes the questionnaire responses if present, and instructs it to create a response for the user that includes a hair care routine, product and ingredient recommendations, and tips for improving hair health.

The code imports the OpenAI API library in order to call built-in functions to retrieve responses.

2. Fine-tuning strategy (are students using pre-trained models, or training their own?)

We are using a pre-trained model, but we are refining its possible responses by explaining what information it is receiving and the layout of the response it should give.

3. Prompt engineering and optimization (how are students structuring prompts for effective responses?)

Some of the questions consist of dropboxes and then the other one is a typed out response allowing the response to be extra user specific. It was important to narrow the variety of

inputs the user could submit. By providing the hair questionnaire, user prompts will exhibit similar builds before continuing on to the hair concerns section. The questionnaire also contributes to honing API responses by supplying information that could otherwise be forgotten if the only input came from a textbox. There is also a specific question for the user to answer within the text box in order to encourage a relevant submission.

# Phase 3 Report

# PHASE 3 REPORT



## Confidence Level

Moderate condition → Balanced recommendations

## Was this recommendation helpful?

Rate this recommendation (1 = Not helpful, 5 = Excellent)



Submit Feedback



# AI Hair Care Recommendation System

Decision Support System Prototype - v4.0

## Hair Assessment Survey

|                                     |                                                        |
|-------------------------------------|--------------------------------------------------------|
| What is your hair type?             | What is your hair density?                             |
| <div>Straight</div>                 | <div>Fine</div>                                        |
| How would you describe your scalp?  | How often do you use heat styling tools?               |
| <div>Dry</div>                      | <div>Never</div>                                       |
| What is your hair porosity?         | Do you use chemical treatments (relaxer, color, etc.)? |
| <div>Low</div>                      | <div>No</div>                                          |
| What is your current hair length?   | How would you rate your current damage level?          |
| <div>Medium (shoulder length)</div> | <div>Low</div>                                         |
| How often do you wash your hair?    | What type of environment do you live in?               |
| <div>Daily</div>                    | <div>Humid</div>                                       |
| What is your main hair goal?        |                                                        |
| <div>Strengthen hair</div>          |                                                        |

What is your main hair goal?

Strengthen hair

What is your most common styling method?

Protective styles (braids, twists, wigs)

How would you describe your water intake and diet?

Healthy diet, drink plenty of water

Tell us more about your hair (optional):

my edges are thin

Get Recommendation

# Your Personalized Recommendation

## Hair Profile Score Dashboard

Generated by Mini Brain rule-based analysis — before AI processing

Moisture Level

Damage Level

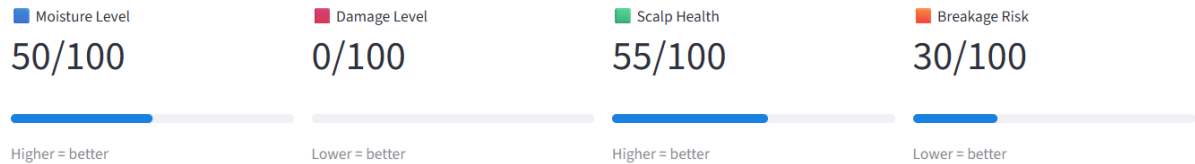
Scalp Health

Breakage Risk

# Your Personalized Recommendation

## Hair Profile Score Dashboard

Generated by Mini Brain rule-based analysis — before AI processing



## Hair Health Score

Score: 6 — Higher = more damage/attention needed

Moderate condition — balanced recommendations

## Detected Issues



 **Detected Issues**

Dry Scalp, Low Porosity, Fine Hair

 **Mini Brain — Detailed Issue Breakdown**

Rule-based analysis completed before AI processing

-  Dry Scalp
-  Low Porosity — Product Buildup Prone
-  Fine Hair — Prone to Breakage
-  Low Damage Level (user-reported)
-  Humid Climate — Frizz Risk

## AI Recommendation

Sure, here's a professional recommendation tailored to your hair type and goals:

### 1. Recommended Shampoo Type: Hydrating Shampoo

Look for shampoos that contain ingredients such as aloe vera, glycerin, and natural oils (like avocado or coconut oil). These ingredients provide moisture without weighing down fine hair, which is essential for low porosity hair that can struggle to absorb hydration. Hydrating shampoos help cleanse the scalp and hair while maintaining moisture levels, which is crucial for your dry scalp and overall hair health.

### 2. Recommended Conditioner Type: Lightweight Leave-In Conditioner

Select a leave-in conditioner formulated with ingredients like shea butter, argan oil, and hydrolyzed proteins. These ingredients provide essential moisture and help to strengthen the hair without causing buildup common with low porosity hair. The lightweight formula is perfect for fine hair, ensuring it doesn't feel heavy or greasy while promoting soft, manageable strands.

### 3. Recommended Treatment: Protein Moisture Treatment

A protein moisture treatment is ideal for your hair as it balances protein and moisture levels, strengthening fine hair prone to breakage. Look for treatments with hydrolyzed proteins and humectants, which can improve the hair's elasticity and strength. Because you mentioned a low damage level, a treatment used sparingly can help maintain and enhance the overall integrity of your hair.

### 4. Suggested Weekly Routine:

- Day 1: Wash hair with hydrating shampoo, follow with a lightweight leave-in conditioner, style in protective braids.
- Day 2: Moisturize scalp and edges lightly with natural oils (like castor or jojoba) to nourish and strengthen.
- Day 3: Apply the protein moisture treatment as directed, focusing on mid-lengths to ends. Rinse thoroughly and follow with a lightweight leave-in.
- Day 4: Rest day; wear hair in a loose protective style and hydrate with a spray of water mixed with a few drops of oil.
- Day 5: Wash hair with hydrating shampoo, condition with a lightweight conditioner, and style as desired.
- Day 6/7: Focus on scalp care—apply a deep moisturizing scalp treatment, and do a gentle scalp massage to promote circulation.

- Day 1: Wash hair with hydrating shampoo, follow with a lightweight leave-in conditioner, style in protective braids.
- Day 2: Moisturize scalp and edges lightly with natural oils (like castor or jojoba) to nourish and strengthen.
- Day 3: Apply the protein moisture treatment as directed, focusing on mid-lengths to ends. Rinse thoroughly and follow with a lightweight leave-in.
- Day 4: Rest day; wear hair in a loose protective style and hydrate with a spray of water mixed with a few drops of oil.
- Day 5: Wash hair with hydrating shampoo, condition with a lightweight conditioner, and style as desired.
- Day 6/7: Focus on scalp care—apply a deep moisturizing scalp treatment, and do a gentle scalp massage to promote circulation.

5. **Explanation for Recommendations: Hair Type: Straight** — Products that provide hydration without heaviness are essential for straight fine hair, allowing for a sleek appearance without flatness.

**Scalp Condition: Dry** — Hydrating products are vital in alleviating dryness, ensuring the scalp remains healthy and fosters hair growth.

**Damage Level: Low** — Since your hair has a low damage level, treatments like the protein moisture treatment should be used sparingly for maintenance without overwhelming your strands.

**Goal: Strengthen hair** — All products are selected to promote moisture retention and strength, addressing both fine texture and low porosity tendencies.

In summary, this tailored hair care regimen will cater to your specific needs and goals, while supporting the health and strength of your hair in a humid environment.

System Confidence Level: 95%

## Confidence Level

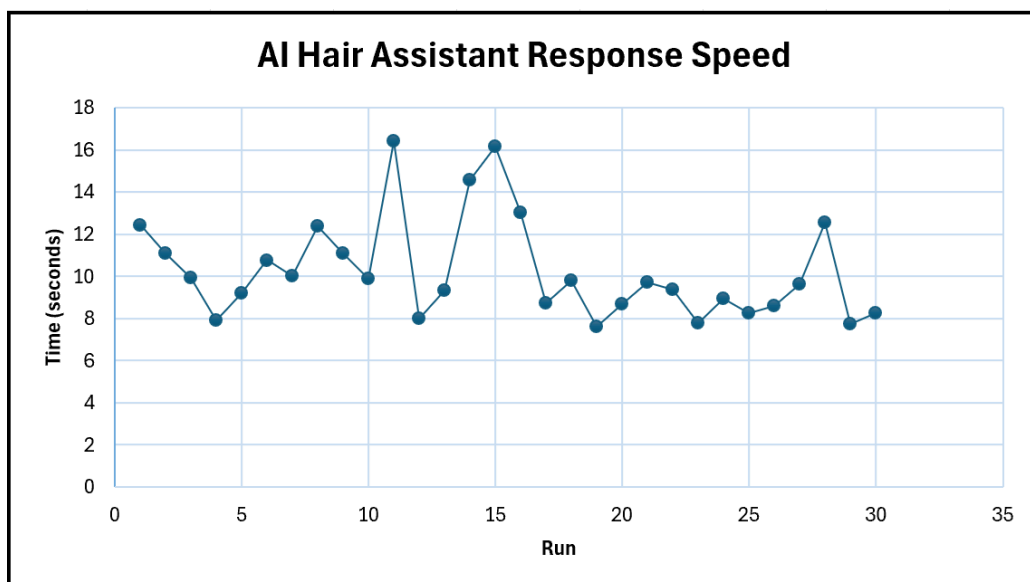
Moderate condition → Balanced recommendations

## Accuracy, Response Time, Usability Testing:

In order to test response time, 30 tests were run on the AI Hair Assistant with response time recorded. Different parameters were applied every time the test was run. It is difficult to gauge accuracy of responses due to there not being a central, reputable source to compare all responses against.

Over the course of 30 tests, the fastest response time was 7.61 seconds and the slowest was 16.441 seconds. The average response time was 10.267 seconds. The following is a table of all recorded responses and a graph to visualize the data.

| RUN | TIME (s) | RUN | TIME (s) |
|-----|----------|-----|----------|
| 1   | 12.45    | 16  | 13.05    |
| 2   | 11.112   | 17  | 8.73     |
| 3   | 9.922    | 18  | 9.801    |
| 4   | 7.9      | 19  | 7.611    |
| 5   | 9.22     | 20  | 8.681    |
| 6   | 10.772   | 21  | 9.722    |
| 7   | 10.021   | 22  | 9.371    |
| 8   | 12.391   | 23  | 7.771    |
| 9   | 11.1     | 24  | 8.931    |
| 10  | 9.88     | 25  | 8.251    |
| 11  | 16.441   | 26  | 8.611    |
| 12  | 7.992    | 27  | 9.621    |
| 13  | 9.33     | 28  | 12.571   |
| 14  | 14.56    | 29  | 7.741    |
| 15  | 16.182   | 30  | 8.252    |



### **Comparative Analysis:**

Our AI Hair Assistant provides a service to users that is typically provided in-person at a hair salon or by a hair specialist, so it is difficult to present a direct comparative analysis. What can be noted is that the AI Hair Assistant does make personalized hair recommendations more accessible for end users who may not have a salon within a reasonable distance of them or those who may not be able to afford a salon's service price.

In contrast to the accessibility, the ability to visit a hair stylist in-person allows for a live demonstration of hair styling techniques that the AI Hair Assistant can only describe. The user may take further steps to find an online video or more in depth description of the technique, but a hair stylist could simply explain the procedure while actively performing it. Another factor to consider in favor of using an in-person hair stylist is that the end user may incorrectly self-report issues about their hair symptoms. There is room in the hair questionnaire for the user to select the incorrect hair type, texture, porosity, and so on which would then cause an irrelevant response. A licensed hair stylist should have a smaller margin of error when recognizing issues and/or qualities about a client's hair.

# AI System Video

<https://drive.google.com/file/d/1Fqz2u9xVQ9YAS1ODEFZOdeCQiZ22qpVr/view>